

Asimptotik Dizi Denklemleri:

bn bir dizi, $l \in \mathbb{R}$ o.c $\lim_{n \rightarrow \infty} a_n \begin{cases} = l \\ = \infty \text{ veya } \emptyset \end{cases} \Rightarrow$ Yakınsak dizi

$S_N = \sum_{n=0}^N$, $N \rightarrow \infty$ $\begin{cases} S_N \text{ bir limite yaklaşıyor yakınsak seri} \\ S_N \text{ limitte yok veya sonsuz giderse ıraksak seri} \end{cases}$

$$\sinh(\varepsilon) = \lim_{\varepsilon \rightarrow 0} \frac{\sinh(\varepsilon)}{\varepsilon} = \lim_{\varepsilon \rightarrow 0} \frac{\varepsilon + \frac{\varepsilon^3}{6} + O(\varepsilon^5)}{\varepsilon} = 1 \Rightarrow \sinh(\varepsilon) = O(\varepsilon)$$

$$\tanh(\varepsilon) = \lim_{\varepsilon \rightarrow 0} \frac{\tanh(\varepsilon)}{\varepsilon} = \lim_{\varepsilon \rightarrow 0} \frac{\varepsilon + \frac{\varepsilon^3}{6} + O(\varepsilon^5)}{\varepsilon} = 1 \Rightarrow \tanh(\varepsilon) = O(\varepsilon)$$

$$\sinh(\varepsilon)^n = \lim_{\varepsilon \rightarrow 0} \frac{(\varepsilon + O(\varepsilon^3))^n}{\varepsilon^n} = \frac{\varepsilon^n + O(\varepsilon^{n+2})}{\varepsilon^n} = 1 \Rightarrow \sinh(\varepsilon)^n = O(\varepsilon^n)$$

$$\sinh(\varepsilon) = O(\cos(\varepsilon)) \Rightarrow \lim_{\varepsilon \rightarrow 0} \frac{\sinh(\varepsilon)}{\cos(\varepsilon)} \approx \lim_{\varepsilon \rightarrow 0} \frac{\varepsilon + O(\varepsilon^3)}{1 + O(\varepsilon^2)} \approx \lim_{\varepsilon \rightarrow 0} \varepsilon = 0$$

$$O(\varepsilon^{-1}) = e^\varepsilon \Rightarrow \lim_{\varepsilon \rightarrow 0} \frac{e^\varepsilon}{\varepsilon^{-1}} = \lim_{\varepsilon \rightarrow 0} \varepsilon \left(1 + \varepsilon + \frac{\varepsilon^2}{2} + \dots\right) = 0$$

$$\sin(\varepsilon) \sim \tan(\varepsilon) \Rightarrow \lim_{\varepsilon \rightarrow 0} \frac{\sin(\varepsilon)}{\tan(\varepsilon)} = \lim_{\varepsilon \rightarrow 0} \cos \varepsilon = 0 \quad (\text{Vize})$$

$$(1-\varepsilon)^{-2} - 1 - 2\varepsilon \sim 3\varepsilon^2 \Rightarrow \lim_{\varepsilon \rightarrow 0} \frac{(1-\varepsilon)^{-2} - 1 - 2\varepsilon}{3\varepsilon^2} = \frac{1+2\varepsilon+3\varepsilon^2+O(\varepsilon^3)-1-2\varepsilon}{3\varepsilon^2}$$

$$\Rightarrow \lim_{\varepsilon \rightarrow 0} \frac{(1+2\varepsilon+3\varepsilon^2+O(\varepsilon^3))-1-2\varepsilon}{3\varepsilon^2} = 0$$

$\Rightarrow x \rightarrow \infty$ için

$$\sinh(x) = O(e^x) \Rightarrow \lim_{x \rightarrow \infty} \frac{\sinh(x)}{e^x} = \lim_{x \rightarrow \infty} \frac{\frac{e^x - e^{-x}}{2}}{e^x} = \lim_{x \rightarrow \infty} \frac{e^x}{2e^x} \approx \frac{1}{2}$$

$$\Rightarrow \lim_{x \rightarrow \infty} \frac{e^x - e^{-x}}{2} \sim \frac{e^x}{2} \text{ (x çok büyükken)}$$

$$x = o(e^x) \Rightarrow \lim_{x \rightarrow \infty} \frac{x}{e^x} \xrightarrow{\text{L'Hopital}} \lim_{x \rightarrow \infty} \frac{1}{e^x} = 0$$

$$x^n = o(e^x) \Rightarrow \lim_{x \rightarrow \infty} \frac{x^n}{e^x} \xrightarrow{\text{n. def. L'Hopital}} \frac{n \cdot (n-1) \cdot (n-2) \cdots x^0}{e^x} = n! \lim_{x \rightarrow \infty} \frac{1}{e^x} = 0$$

$$e^{-x} = o(x^{-n}) \Rightarrow \lim_{x \rightarrow \infty} \frac{e^{-x}}{x^{-n}} = \lim_{x \rightarrow \infty} \frac{x^n}{e^x} \xrightarrow{\text{n. def. L'Hopital}} n! \lim_{x \rightarrow \infty} \frac{1}{e^x} = 0$$

$$e^{-\frac{x}{2}} = o(x^{-n}) \Rightarrow \lim_{x \rightarrow \infty} \frac{e^{-\frac{x}{2}}}{x^{-n}} = \lim_{x \rightarrow \infty} \frac{x^n}{e^{\frac{x}{2}}} \xrightarrow{\text{n. def. L'Hopital}} n! \cdot 2^n \lim_{x \rightarrow \infty} \frac{1}{e^{\frac{x}{2}}} = 0$$

$\Rightarrow \varepsilon \rightarrow 0$ için

$$e^{-\varepsilon} = \lim_{\varepsilon \rightarrow 0} e^{-\varepsilon} = \lim_{\varepsilon \rightarrow 0} \left(1 - \varepsilon + \frac{\varepsilon^2}{2} + O(\varepsilon^3) \right) = 1 \Rightarrow e^{-\varepsilon} = O(1)$$

$$\cot(\varepsilon) = \frac{\cos(\varepsilon)}{\sin(\varepsilon)} = \lim_{\varepsilon \rightarrow 0} \frac{1 - \frac{\varepsilon^2}{2} + O(\varepsilon^4)}{\varepsilon - \frac{\varepsilon^3}{6} + O(\varepsilon^5)} \approx \frac{1}{\varepsilon} \Rightarrow \cot(\varepsilon) = O(\varepsilon^{-1})$$

$$\ln(1+\varepsilon) \sim \varepsilon \Rightarrow \ln(1+\varepsilon) = O(\varepsilon) \Rightarrow \lim_{\varepsilon \rightarrow 0} \frac{\varepsilon - \frac{\varepsilon^2}{2} + \frac{\varepsilon^3}{3}}{\varepsilon} = 1$$

$$\frac{1 - \cos(\varepsilon)}{1 + \cos(\varepsilon)} = \lim_{\varepsilon \rightarrow 0} \frac{1 - (1 - \frac{1}{2}\varepsilon^2 + \frac{1}{24}\varepsilon^4 - \dots)}{1 + (1 - \frac{1}{2}\varepsilon^2 + \frac{1}{24}\varepsilon^4 - \dots)} = \lim_{\varepsilon \rightarrow 0} \frac{\frac{1}{2}\varepsilon^2 + O(\varepsilon^4)}{2 + O(\varepsilon^2)} = \frac{\varepsilon^2}{4} \Rightarrow \frac{1 - \cos(\varepsilon)}{1 + \cos(\varepsilon)} = O(\varepsilon^2)$$

$$\sinh\left(\frac{1}{\varepsilon}\right) = \lim_{\varepsilon \rightarrow 0} \frac{e^{\frac{1}{\varepsilon}} - e^{-\frac{1}{\varepsilon}}}{2} \sim \frac{e^{\frac{1}{\varepsilon}}}{2} \Rightarrow \sinh\left(\frac{1}{\varepsilon}\right) = O(e^{\frac{1}{\varepsilon}})$$

$$\cos(\pi x) \sim \pi \left(\frac{1}{2} - x\right), \quad x = \frac{1}{2}$$

$$\lim_{x \rightarrow \frac{1}{2}} \frac{\cos(\pi x)}{\pi \left(\frac{1}{2} - x\right)} \xrightarrow{\text{L'Hospital}} \lim_{x \rightarrow \frac{1}{2}} \frac{-\pi \sin(\pi x)}{-\pi} = 1 \neq 0$$

$$e^{-\frac{1}{x}} = o(x^n), \quad x \rightarrow 0^+$$

$$\lim_{x \rightarrow 0^+} \frac{e^{-\frac{1}{x}}}{x^n} \xrightarrow[\substack{x=t \\ t \rightarrow 0}]{\text{L'Hospital}} \lim_{t \rightarrow 0} \frac{e^{-t}}{t^{-n}} = t! \lim_{t \rightarrow 0} e^{-t} = 0$$

$$x^{-n} = o(e^{-\frac{1}{x}}), \quad x \rightarrow 0^+$$

$$\lim_{x \rightarrow 0^+} \frac{x^{-n}}{e^{-\frac{1}{x}}} \xrightarrow[\substack{-\frac{1}{x}=t \\ t \rightarrow 0}]{\text{L'Hospital}} \lim_{t \rightarrow 0} \frac{\left(-\frac{1}{t}\right)^{-n}}{e^t} = \lim_{t \rightarrow 0} \frac{t^n}{e^t} = 0$$

OP3: Bestimme den kleinen ε entsprechenden asymptotische Entwicklung
für die Funktion $\varepsilon \rightarrow 0$ sein lassen.

$$2-1) x^2 + x - \varepsilon = 0, \quad x = x_0 + \varepsilon x_1 + o(\varepsilon^2)$$

$$(x_0 + \varepsilon x_1)^2 + (x_0 + \varepsilon x_1) - \varepsilon = 0$$

$$x_0^2 + 2\varepsilon x_0 x_1 + \varepsilon^2 x_1^2 + x_0 + \varepsilon x_1 - \varepsilon = 0$$

0(1) termlen:

$$x_0^2 + x_0 = 0$$

$$x_0(x_0 + 1) = 0 \Rightarrow x_0 = 0, x_0 = -1$$

I. Ansatz, $x_0 = 0$

$$2(0) x_1 + x_1 - 1 = 0 \Rightarrow x_1 = 1 \Rightarrow x = 0 + \varepsilon + o(\varepsilon^2)$$

$$\Rightarrow x \sim \varepsilon$$

II. Ansatz, $x_0 = -1$

$$2(-1) x_1 + x_1 - 1 = 0 \Rightarrow x_1 = -1 \Rightarrow x = -1 - \varepsilon + o(\varepsilon^2)$$

$$\Rightarrow x \sim -1 - \varepsilon$$

$$b-1) x^3 + \varepsilon x + 1 = 0$$

$$(x_0 + \varepsilon x_1)^3 + \varepsilon(x_0 + \varepsilon x_1) + 1 = x_0^3 + 1 + \varepsilon(3x_0^2 x_1 + x_0) = 0$$

$$0(1) \text{ termlen: } x_0^3 + 1 = 0 \Rightarrow x_0 = -1$$

$$0(\varepsilon) \text{ term: } 3(-1)^2 x_1 - 1 = 0$$

$$3x_1 - 1 = 0$$

$$x_1 = \frac{1}{3} \Rightarrow x = -1 + \frac{\varepsilon}{3} + o(\varepsilon^2)$$

$$x \sim -1 + \frac{\varepsilon}{3}$$

$$C-1) x^3 + \varepsilon x^2 - 4x = 0, \quad x = x_0 + \varepsilon x_1 + o(\varepsilon^2)$$

$$(x_0 + \varepsilon x_1)^3 + \varepsilon (x_0 + \varepsilon x_1)^2 - 4(x_0 + \varepsilon x_1) = 0$$

$$x_0^3 + 3\varepsilon x_0^2 x_1 + o(\varepsilon^2) + \varepsilon x_0^2 + o(\varepsilon^2) - 4x_0 - 4\varepsilon x_1 = 0$$

$$x_0^3 - 4x_0 + \varepsilon (3x_0^2 x_1 + x_0^2 - 4x_1) = 0$$

0(1) term;

$$x_0^3 - 4x_0 = 0$$

$$x_0^2 (x_0 - 4) = 0$$

$$x_0 = 0$$

$$x_0 = 2$$

$$x_0 = -2$$

I. Surun $x_0 = 0$

$$-4\varepsilon x_1 = 0 \Rightarrow x_1 = 0 \Rightarrow x \sim 0$$

II. Surun $x_0 = 2$

$$\left. \begin{aligned} 8 - 8 + \varepsilon (12x_1 + 4 - 4x_1) &= 0 \\ 8x_1 &= -4 \Rightarrow x_1 = -\frac{1}{2} \end{aligned} \right\} \begin{aligned} x &= 2 - \frac{\varepsilon}{2} + o(\varepsilon^2) \\ x &\sim 2 - \frac{\varepsilon}{2} \end{aligned}$$

III. Surun $x_0 = -2$

$$\left. \begin{aligned} -8 + 8 + \varepsilon (12x_1 + 4 - 4x_1) &= 0 \\ 8x_1 &= -4 \Rightarrow x_1 = -\frac{1}{2} \end{aligned} \right\} \begin{aligned} x &= -2 - \frac{\varepsilon}{2} + o(\varepsilon^2) \\ x &\sim -2 - \frac{\varepsilon}{2} \end{aligned}$$

$$x = o\left(\frac{\varepsilon}{2}\right)$$

~~Ör~~ $\varepsilon x^2 + 2x - 1 = 0$ denk. $\varepsilon \ll 1$ için iki köküne bakalım

$\varepsilon = 0$ için $2x - 1 = 0 \Rightarrow x = \frac{1}{2}$ yani tek köküne sahip ama

$0 < \varepsilon \ll 1$ için iki köküne sahip ilk önce kökümüzü

$$x = x_0 + \varepsilon x_1 + O(\varepsilon^2)$$

formülde yazalım

$$\varepsilon (x_0 + \varepsilon x_1)^2 + 2(x_0 + \varepsilon x_1) - 1 + O(\varepsilon^2) = 0$$

$$\varepsilon x_0^2 + O(\varepsilon^2) + 2x_0 + 2\varepsilon x_1 - 1 + O(\varepsilon^2) = 0$$

0.11'de: $2x_0 - 1 = 0 \Rightarrow x_0 = \frac{1}{2}$

0.11'de

$$x_0^2 + 2x_1 = 0 \Rightarrow \left(\frac{1}{2}\right)^2 + 2x_1 = 0 \Rightarrow x_1 = -\frac{1}{8}$$

$\therefore x = x_0 + \varepsilon x_1 + O(\varepsilon^2) = \frac{1}{2} - \frac{\varepsilon}{8} + O(\varepsilon^2)$ yazarsak tek kök buluruz. 0 halde

kelecekte kök $|x| \gg 1$ olacaktır

$$x = \varepsilon^{-\alpha} \mathcal{X}, \quad \mathcal{X} = O(1) \quad \text{ve} \quad \alpha > 0$$

seçelim ve denk'de yerine yazalım

$$\varepsilon (\varepsilon^{-\alpha} \mathcal{X})^2 + 2\varepsilon^{-\alpha} \mathcal{X} - 1 = 0$$

e

$$\underbrace{\varepsilon^{1-2\alpha} \mathcal{X}^2}_{O(\varepsilon^{1-2\alpha})} + \underbrace{2\varepsilon^{-\alpha} \mathcal{X}}_{O(\varepsilon^{-\alpha})} - \underbrace{1}_{O(1)} = 0$$

* $O(\varepsilon^{1-2\alpha}) = O(\varepsilon^{-\alpha})$, birbirlerine eşit olsun

$$1-2\alpha = -\alpha$$

$$\alpha = 1$$

, denkleme yerine yaz

$$* x = \varepsilon^{-\alpha} \mathcal{X}$$

$$= \varepsilon^{-1} \mathcal{X}$$

$$= \frac{\mathcal{X}}{\varepsilon}$$

Test Bidden $\rightarrow \varepsilon^{-1}$ da kontinuität

$$\varepsilon^{-1} x^2 + 2\varepsilon^{-1} x - 1 = 0$$

$$x^2 + 2x - \varepsilon = 0$$

Sind $x_0 = \varepsilon x_1 + O(\varepsilon^2)$ die dazugehörige gen. yz sein

$$(x_0 + \varepsilon x_1)^2 + 2(x_0 + \varepsilon x_1) - \varepsilon = 0$$

$$x_0^2 + 2\varepsilon x_0 x_1 + O(\varepsilon^2) + 2x_0 + 2\varepsilon x_1 - \varepsilon = 0$$

0(1)de

$$x_0^2 + 2x_0 = 0 \Rightarrow x_0 = -2, x_0 = 0$$

1. Ann: $x_0 = 0$ voraus $O(\varepsilon)$ de

$$2x_0 x_1 + 2x_1 - 1 = 0$$

$$0 + 2x_1 - 1 = 0$$

$x_1 = \frac{1}{2} \Rightarrow$ bis in alle betr. h. gültig sein
 $x = \frac{1}{2} + O(\varepsilon)$

2. Ann: $x_0 = -2$ voraus $O(\varepsilon)$ de

$$2x_0 x_1 + 2x_1 - 1 = 0$$

$$-4x_1 + 2x_1 - 1 = 0$$

$$x_1 = -\frac{1}{2}$$

$$x_1 = \frac{x_1}{\varepsilon} = -\frac{1}{2\varepsilon} \quad \text{also}$$

$$x_0 = \frac{x_0}{\varepsilon} = -\frac{2}{\varepsilon}$$

$$x = x_0 + \varepsilon x_1 + O(\varepsilon^2)$$

$$= -\frac{2}{\varepsilon} - \varepsilon \frac{1}{2\varepsilon} + O(\varepsilon^2)$$

$$= -\frac{2}{\varepsilon} - \frac{1}{2} + O(\varepsilon^2)$$

$$\epsilon x^4 - x^2 - x + 2 = 0$$

$$\epsilon (x_0 + \epsilon x_1 + o(\epsilon^2))^4 - (x_0 + \epsilon x_1 + o(\epsilon^2))^2 - (x_0 + \epsilon x_1 + o(\epsilon^2)) + 2 = 0$$

$$\epsilon x_0^4 + o(\epsilon^2) - (x_0^2 - 2\epsilon x_0 x_1 + o(\epsilon^2)) - x_0 + \epsilon x_1 + o(\epsilon^2) + 2 = 0$$

$$-(x_0^2 - x_0 + 2) + \epsilon (x_0^4 - 2x_0 x_1 - x_1) = 0$$

$O(1)$ terms:

$$-(x_0^2 - x_0 + 2) = 0 \Rightarrow x_0 = 1, x_0 = -2$$

I. Durum $x_0 = 1$

$$-1 - 1 + 2 + \epsilon (1 - 2x_1 - x_1) = 0 \Rightarrow x_1 = \frac{1}{3}$$

$$\Rightarrow x = 1 + \frac{\epsilon}{3} + o(\epsilon^2) \Rightarrow x \sim 1 + \frac{\epsilon}{3}$$

II. Durum $x_0 = -2$

$$-4 + 2 + 2 + \epsilon (16 + 4x_1 - x_1) = 0 \Rightarrow x_1 = -\frac{16}{3}$$

$$\Rightarrow x = -2 - \frac{16\epsilon}{3} + o(\epsilon^2) \Rightarrow 1 + \frac{8\epsilon}{3} \approx x$$

$$x = \epsilon^\alpha x \Rightarrow \epsilon \epsilon^{-4\alpha} x^4 - \epsilon^{-2\alpha} x^2 - \epsilon^{-\alpha} x + 2 = 0 \Rightarrow 1 - 4\alpha = -2\alpha$$

$$\alpha = \frac{1}{2}$$

$$\Rightarrow x^4 - x^2 - \epsilon^{\frac{1}{2}} x + 2\epsilon = 0 \Rightarrow$$

$$\Rightarrow x_0^4 - x_0^2 = 0 \Rightarrow x_0 = 0, \pm 1$$

I. Durum $x_0 = 0$ $\Rightarrow$

II. Durum $x_0 = 1$ $\Rightarrow$

III. Durum $x_0 = -1$ $\Rightarrow$

$$\epsilon - \epsilon^3 x^2 + \epsilon x + 1 = 0, \quad x = \epsilon^\alpha x$$

$$\underbrace{\epsilon^3 (\epsilon^{-2\alpha} x^2)}_{O(\epsilon^{3-2\alpha})} + \underbrace{\epsilon (\epsilon^{-\alpha} x)}_{O(\epsilon^{1-\alpha})} + 1 = 0$$

$$\Rightarrow 3-2\alpha = 1-\alpha$$

$$\alpha = 2$$

$$\epsilon^{-2} x^2 + \epsilon^{-1} x + 1 = 0$$

Ans. further $\Rightarrow x^2 + x + \epsilon = 0$

$$x_0^2 + 2\epsilon x_0 x_1 + x_0 + \epsilon x_1 + \epsilon = 0$$

$O(1)$ term

$$x_0^2 + x_0 = 0 \Rightarrow x_0 = 0, x_0 = -1$$

$x_0 = 0$ sein $O(\epsilon)$

$$x_1 + 1 = 0 \Rightarrow x_1 = -1 \Rightarrow x = -\epsilon + O(\epsilon^2)$$

$x_0 = -1$ sein $O(\epsilon)$

$$-2x_1 + x_1 + 1 = 0 \Rightarrow x_1 = 1 \Rightarrow x = -1 + \epsilon + O(\epsilon^2)$$

✓ Q2

$$1) \quad \varepsilon^2 x^3 + x^2 + 2x + \varepsilon = 0$$

$$\underbrace{0(\varepsilon^2)} + \underbrace{x_0^2} + 2\varepsilon x_0 x_1 + o(\varepsilon^2) + 2(x_0 + \varepsilon x_1 + o(\varepsilon^2)) + \varepsilon = 0$$

$$x_0^2 + 2x_0 + \varepsilon(2x_0 x_1 + 2x_1 + 1) = 0$$

$O(1)$ termi;

$$x_0^2 + 2x_0 = 0$$

$$x_0(x_0 + 2) = 0 \quad \begin{cases} x_0 = -2 \\ x_0 = 0 \end{cases}$$

I. Durum $x_0 = 0$

$$0 + 0 + \varepsilon(0 + 2x_1 + 1) = 0$$

$$2x_1 = -1$$

$$x_1 = -\frac{1}{2}$$

$$\Rightarrow x = 0 - \frac{\varepsilon}{2} + o(\varepsilon^2)$$

$$x \sim -\frac{\varepsilon}{2}$$

II. Durum $x_0 = -2$

$$4 - 4 + \varepsilon(-4x_1 + 2x_1 + 1) = 0$$

$$-2x_1 = -1$$

$$x_1 = \frac{1}{2}$$

$$\Rightarrow x = -2 + \frac{\varepsilon}{2} + o(\varepsilon^2)$$

$$x \sim -2 + \frac{\varepsilon}{2}$$

$$f.) \varepsilon^2 x^3 + x^2 + 2x + \varepsilon = 0 \quad \text{suchen}$$

$$x = \varepsilon^{-\lambda} \tilde{x}$$

$$\varepsilon^2 (\varepsilon^{-3\lambda} \tilde{x}^3) + \varepsilon^{2\lambda} \tilde{x}^2 + 2\varepsilon^{-\lambda} \tilde{x} + \varepsilon = 0$$

$$O(\varepsilon^{2-3\lambda}) \quad O(\varepsilon^{2\lambda}) \quad O(\varepsilon^{-\lambda}) \Rightarrow 2-3\lambda = -2\lambda$$

$$\lambda = 2$$

$$\varepsilon^{-4} \tilde{x}^3 + \varepsilon^{-4} \tilde{x}^2 + 2\varepsilon^{-2} \tilde{x} + \varepsilon = 0$$

$$\text{New Looken } \tilde{x}^3 + \tilde{x}^2 + 2\varepsilon^2 \tilde{x} + \varepsilon^5 = 0$$

$$\Rightarrow \tilde{x}_0^3 + 2\varepsilon \tilde{x}_0^2 \tilde{x}_1 + \tilde{x}_0^2 + 2\varepsilon \tilde{x}_0 \tilde{x}_1 = 0$$

O(1)

$$\tilde{x}_0^3 + \tilde{x}_0^2 = 0 \Rightarrow \tilde{x}_{1,2} = 0 \quad \tilde{x}_3 = -1$$

$$\tilde{x}_0 = 0 \quad \text{kein } O(\varepsilon) \text{ da } \tilde{x}_1 = 0$$

$$\tilde{x}_0 = -1 \quad \text{kein } O(\varepsilon)$$

$$2\varepsilon \tilde{x}_1 - 2\varepsilon \tilde{x}_1 = 0$$

$$\epsilon x^2 + e^{\epsilon x} - x = 0 \quad (\text{bulunur ihtimali yüksek})$$

Böylece $0 < \epsilon \ll 1$ için iki tarafta da asimptotik olarak aynı terimleri kullanarak

$$e^{\epsilon x} = 1 + \epsilon x_0 + \epsilon^2 \left(x_1 + \frac{x_0^2}{2} \right) + O(\epsilon^3)$$

Böylece genelleştirilir

$$\epsilon \left(x_0 + \epsilon x_1 + O(\epsilon^2) \right)^2 + (1 + \epsilon x_0 + O(\epsilon^2)) - (x_0 + \epsilon x_1 + O(\epsilon^2)) = 0$$

$$1 - x_0 + \epsilon(x_0^2 + x_0 - x_1) = 0$$

$O(1)$ terimi!

$$1 - x_0 = 0$$

$$x_0 = 1$$

$O(\epsilon)$ terimi!

$$1 + 1 - x_1 = 0$$

$$x_1 = 2$$

$$x = x_0 + \epsilon x_1 + O(\epsilon^2) = 1 + 2\epsilon + O(\epsilon^2)$$

$$x \sim 1 + 2\epsilon$$

Finalde
kesir
Bek

$$X = \varepsilon^{-\alpha} \chi$$

$$\underbrace{\varepsilon |\varepsilon^{-2\alpha} \chi^2|}_{O(\varepsilon^{1-2\alpha})} + \underbrace{e^{\varepsilon \varepsilon^{-\alpha} \chi} - \varepsilon^{-\alpha} \chi}_{O(\varepsilon^{-\alpha})} = 0 \Rightarrow 1-2\alpha = -\alpha$$

$$\alpha = 1$$

$$\varepsilon^{-1} \chi^2 + e^{\chi} - \varepsilon^{-1} \chi = 0$$

$$\chi^2 - \chi + \varepsilon e^{\chi} = 0$$

$$\chi_0^2 + 2\varepsilon \chi_0 \chi_1 - \chi_0 + \varepsilon \chi_1 + \varepsilon(1 + \chi_0) = 0$$

O(1)

$$\chi_0^2 - \chi_0 = 0 \Rightarrow \chi_0 = 0, \chi_0 = 1$$

$\chi_0 = 0$ with $O(\varepsilon)$

$$-\chi_1 + 1 = 0 \Rightarrow \chi_1 = 1 \Rightarrow \chi = \varepsilon + O(\varepsilon^2)$$

$\chi_0 = 1$ with $O(\varepsilon)$

$$2\chi_1 - \chi_1 + 2 = 0 \Rightarrow \chi_1 = -2 \Rightarrow \chi = 1 - 2\varepsilon + O(\varepsilon^2)$$

Örnek $\varepsilon \frac{d^2 y}{dx^2} + \frac{dy}{dx} = 3x^2, y(0)=0, y(1)=2$ sınır değer prob. de çözüm!

$$\varepsilon \left(\frac{d^2 y_0}{dx^2} + \varepsilon \frac{d^2 y_1}{dx^2} + O(\varepsilon^2) \right) + \left(\frac{dy_0}{dx} + \varepsilon \frac{dy_1}{dx} + O(\varepsilon^2) \right) = 3x^2$$

$\varepsilon=0$ yazarız Def. Difer. denklemlerinden ilkinde birine dğer,

Ölkü $\frac{dy_0}{dx} = 3x^2 \Rightarrow y_0 = x^3 + A_0 \Rightarrow y_0(0) = 0^3 + A_0 = 0 \Rightarrow A_0 = 0$
 $y_0(1) = 1^3 + A_0 = 2 \Rightarrow A_0 = 1 \neq$ çözümlü
 $y_0'(x) = 3x^2$
 $y_0''(x) = 6x$

Ölkü $\frac{d^2 y_0}{dx^2} + \frac{dy_1}{dx} = 0 \Rightarrow \frac{dy_1}{dx} = -\frac{d^2 y_0}{dx^2} \Rightarrow \frac{dy_1}{dx} = -6x \Rightarrow y_1(x) = -3x^2 + A_1$

$$y_0(0) = -3x^2 + A_1 = 0 \Rightarrow A_1 = 0$$

$$y_1(1) = -3x^2 + A_1 = 0 \Rightarrow A_1 = 3 \neq$$
 çözümlü

$x=1$ deki süreklilik sınır koşullarını kabul ediyoruz

$$\therefore y(x) = 1 + x^3 + 3\varepsilon(1-x^2) + O(\varepsilon^2)$$

Birde bunu diğ. gösterim deire

$$x = \varepsilon^\alpha \chi, \alpha > 0, \chi = O(1), \varepsilon \rightarrow 0$$

$$y = Y(\chi), Y = O(1), \varepsilon \rightarrow 0$$

$$\frac{dy}{dx} = \frac{dY}{d\chi} \frac{d\chi}{dx} = \frac{dY}{d\chi} \varepsilon^{-\alpha}$$

$$\frac{d^2 y}{dx^2} = \frac{d}{dx} \left(\frac{dY}{d\chi} \varepsilon^{-\alpha} \right) = \frac{d\chi}{dx} \frac{d}{d\chi} \left(\frac{dY}{d\chi} \varepsilon^{-\alpha} \right) = \varepsilon^{-2\alpha} \frac{d^2 Y}{d\chi^2}$$

α zindü deklende yemne koğzobllırız.

$$\varepsilon \varepsilon^{-2\alpha} \frac{d^2 Y}{dx^2} + \varepsilon^{-\alpha} \frac{dY}{dx} = 3(\varepsilon^\alpha x)^2$$

$$\underbrace{\varepsilon^{1-2\alpha} \frac{d^2 Y}{dx^2}}_{O(\varepsilon^{1-2\alpha})} + \underbrace{\varepsilon^{-\alpha} \frac{dY}{dx}}_{O(\varepsilon^{-\alpha})} = \underbrace{3 \varepsilon^{2\alpha} x^2}_{O(\varepsilon^{2\alpha})}$$

$$1-2\alpha = -\alpha$$

$$\alpha = 1 \Rightarrow x = \varepsilon^\alpha x = \varepsilon x, \text{ better choice}$$

$$\varepsilon^{-1} \frac{d^2 Y}{dx^2} + \varepsilon^{-1} \frac{dY}{dx} = 3 \varepsilon^2 x^2$$

$$\frac{d^2 Y}{dx^2} + \frac{dY}{dx} = 3 \varepsilon^3 x^2, \quad y(0)=0 \Rightarrow Y(0/\varepsilon)=0, \quad y(1)=2 \Rightarrow Y(1/\varepsilon)=2 \quad (\text{---})$$

$$\left| \frac{d^2 y_0}{dx^2} + \varepsilon \frac{d^2 y_1}{dx^2} \right| + \left| \frac{dy_0}{dx} + \varepsilon \frac{dy_1}{dx} \right| = 0$$

$$O(1) \text{de} \quad \frac{d^2 y_0}{dx^2} + \frac{dy_0}{dx} = 0 \Rightarrow \text{homogeneous DE} \quad m^2 + m = 0 \Rightarrow m=0 \quad m=-1$$

$$\text{Genl. case: } y_0(x) = A_0 + B_0 e^{-x}$$

$$y_0(0) = A_0 + B_0 = 0 \Rightarrow B_0 = -A_0$$

$$\text{Boundary: } y_0(x) = A_0 - A_0 e^{-x} = A_0 (1 - e^{-x})$$

$$O(\varepsilon) \text{de} \quad \frac{d^2 y_1}{dx^2} + \frac{dy_1}{dx} = 0$$

$$\left\{ \begin{array}{l} \text{Boundary is given} \\ y(x) \sim 1 - e^{-x} \end{array} \right.$$

$$y(x) = 1 + x^3 + 3\varepsilon(1-x^2) + O(\varepsilon^2)$$

$$\text{Again, solve } y_1(x) = A_1(1 - e^{-x}) \quad \text{boundary } 0 + 1 - e^{-x/\varepsilon} + \varepsilon A_1(1 - e^{-x/\varepsilon}) + O(\varepsilon^2)$$

$$\text{Dir. case} = \text{in case}$$

$$\lim_{x \rightarrow 0} y(x) = \lim_{x \rightarrow 0} Y(x)$$

$$\lim_{x \rightarrow 0} (1+x^3) = \lim_{x \rightarrow 0} A_0(1 - e^{-x})$$

$$1 - A_0$$

Asymptotik
Ergänzung

Asimptotik Eşleştirme Arzı Değişkenleri

$0 < \epsilon \ll x \ll 1$, $\epsilon \ll \epsilon^\beta \ll 1$ bir $\hat{x}$ az değişkenlere kıyasla

$$x = \epsilon^\beta \hat{x}, \quad \hat{x} = O(1), \quad 0 < \beta < 1$$

Bir Çözüm

$$y = 1 + x^3 + 3\epsilon(1 - x^2) + O(\epsilon^2)$$

$$= 1 + (\epsilon^\beta \hat{x})^3 + 3\epsilon(1 - (\epsilon^\beta \hat{x})^2) + O(\epsilon^2)$$

Yeni Çözüm

$$x = \epsilon^\beta \hat{x}, \quad \chi = \frac{x}{\epsilon} = \epsilon^{\beta-1} \hat{x}$$

$$y = A_0(1 - e^{-\chi}) + \epsilon A_1(1 - e^{-\chi}) + O(\epsilon^2)$$

$$= A_0(1 - e^{-\epsilon^{\beta-1} \hat{x}}) + \epsilon A_1(1 - e^{-\epsilon^{\beta-1} \hat{x}}) + O(\epsilon^2)$$

Ancak $\epsilon^{\beta-1} \gg 1$ ve $\hat{x} > 0$ ve $\hat{x} = O(1)$

Dolayısıyla $e^{-\epsilon^{\beta-1} \hat{x}} \ll \epsilon^2$, $\epsilon \rightarrow 0$ için

$$y = 1 + \epsilon^{3\beta} \hat{x}^3 + 3\epsilon - 3\epsilon^{1+2\beta} \hat{x}^2 + O(\epsilon^2)$$

$$y = A_0 + \epsilon A_1 + O(\epsilon^2)$$

Bu ilişki eşleştirme uygundur. $A_0 = 1$, $A_1 = 3$ bulunur.

$$y(x) \sim (1 - e^{-x}) + 3\epsilon(1 - e^{-x})$$

Asymptotische Expansions / Van Dyke

$x = \varepsilon X$ einsetzen in Dgl. lösen

$$y^{(1,1)} = 1 + (\varepsilon X)^3 + 3\varepsilon(1 - (\varepsilon X)^2) + O(\varepsilon^2) \\ = 1 + 3\varepsilon + O(\varepsilon^2)$$

9a. abzuw. $x = \frac{X}{\varepsilon}$ einsetzen

$$y^{(1,1)} = A_0 (1 - e^{-\frac{X}{\varepsilon}}) + \varepsilon A_1 (1 - e^{-\frac{X}{\varepsilon}}) + O(\varepsilon^2) \\ = A_0 + \varepsilon A_1 + O(\varepsilon^2)$$

Van Dyke'sche $\Rightarrow 1 + 3\varepsilon = A_0 + \varepsilon A_1, A_0 = 1, A_1 = 3$

Zweite Grenzform $y_c^{(N,M)} = y_{\text{Gleich}} + y_{\text{Gleich}} - \text{Korrekturen (erhöhten / Korr.)}$

$$= 1 - e^{-X} + 3\varepsilon(1 - e^{-X}) + 1 + X^3 + 3\varepsilon(1 - X^2) - (1 + 3\varepsilon) + o(\varepsilon)$$

$$= 1 - e^{-\frac{X}{\varepsilon}} + 3\varepsilon(1 - e^{-\frac{X}{\varepsilon}}) + 1 + X^3 + 3\varepsilon(1 - X^2) - 1 - 3\varepsilon + o(\varepsilon)$$

$$= X^3 - 3\varepsilon X^2 + 3\varepsilon X^2 + (1 + 3\varepsilon)(1 - e^{-\frac{X}{\varepsilon}}) + o(\varepsilon)$$

$$y_c(0) = 0 + o(\varepsilon)$$

$$y_c(1) = 1 - 3\varepsilon + (1 + 3\varepsilon)(1 - e^{-\frac{1}{\varepsilon}}) + o(\varepsilon)$$

$\hookrightarrow e^{-\frac{1}{\varepsilon}} \ll \varepsilon^N, N > 0 \text{ in } \varepsilon \rightarrow 0$ 9a.11

$$= 1 - 3\varepsilon + 1 + 3\varepsilon + o(\varepsilon)$$

$$= o(\varepsilon)$$

$$y_{\text{Gleich}} = X^3 - 3\varepsilon X^2 + 6\varepsilon^2 X + (1 + 3\varepsilon - 6\varepsilon^2) \left(\frac{1 - e^{-\frac{X}{\varepsilon}}}{1 - e^{-\frac{1}{\varepsilon}}} \right)$$

2) $y(x)$ fonksiyonu aşağıdaki dif. den. sağladığını göre

$$\frac{d^2 y}{dx^2} + 2\varepsilon \frac{dy}{dx} + y = 0, \quad y(0) = 0, \quad y(1) = 1, \quad 0 \leq x \leq 1$$

Dif. den. ε çok küçük ilk iki termini aşağıdaki formda bul

$$\left. \begin{aligned} y(x) &= y_0(x) + \varepsilon y_1(x) + O(\varepsilon^2) \\ \frac{dy}{dx} &= \frac{dy_0(x)}{dx} + \varepsilon \frac{dy_1(x)}{dx} + O(\varepsilon^2) \\ \frac{d^2 y}{dx^2} &= \frac{d^2 y_0(x)}{dx^2} + \varepsilon \frac{d^2 y_1(x)}{dx^2} + O(\varepsilon^2) \end{aligned} \right\} \begin{array}{l} 0 < \varepsilon \ll 1 \\ \text{denklemleri yerine yaz} \end{array}$$

$$\left(\frac{d^2 y_0(x)}{dx^2} + \varepsilon \frac{d^2 y_1(x)}{dx^2} \right) + 2\varepsilon \left(\frac{dy_0(x)}{dx} + \varepsilon \frac{dy_1(x)}{dx} \right) + (y_0(x) + \varepsilon y_1(x) + O(\varepsilon^2)) = 0$$

0(1) terimi: $\frac{d^2 y_0}{dx^2} + y_0 = 0$

$y_0(x)$ için: $r^2 + 1 = 0$

Çözüm: $r = \pm i$

Genel çöz: $y_0(x) = A \cos(x) + B \sin(x)$

S.k:

$y_0(0) = 0 \Rightarrow A = 0 \Rightarrow y_0(x) = B \sin(x)$
 $y_0'(x) = B \cos(x)$

$y_0(1) = 1 \Rightarrow B = 1$

0(ε) terimi: $\frac{d^2 y_1}{dx^2} + 2 \frac{dy_0}{dx} + y_1 = 0$

$y_1(x)$ için: $\frac{d^2 y_1}{dx^2} + y_1 = -2 \frac{dy_0}{dx}$

$\Rightarrow \frac{d^2 y_1}{dx^2} + y_1 = -2B \cos(x)$

Homogen hnt çöz

$y_{1h}(x) = C \cos(x) + D \sin(x)$

Özel çözüm

$y_{1p} = E \cos(x)$

$y_{1p}' = -E \sin(x)$

$y_{1p}'' = -E \cos(x)$

denklemleri
yerine yaz

$$(-E \cos(x)) + E \cos(x) = -2B \cos(x)$$

$$0 = -2B \cos(x)$$

Bu özel çözüm gidiyor. Yeni $x \sin(x)$ ve $x \cos(x)$ ikisini bir terim kılacak
Yeni özel çözü

$$y_{ip}(x) = x (P \cos(x) + Q \sin(x))$$

$$y'_{ip}(x) = P \cos(x) + Q \sin(x) - x (P \sin(x) - Q \cos(x))$$

$$y''_{ip}(x) = -P \sin(x) + Q \cos(x) - P \sin(x) + Q \cos(x) - x (P \cos(x) + Q \sin(x))$$

Özel çözüm $y_{ip}(x) = Bx \sin(x)$ dir

$$\text{Genel çözüm : } y_1(x) = C \cos(x) + D \sin(x) + Bx \sin(x)$$

$$\text{Sıfır: Tam çözüme geçelim } y(x) = y_0(x) + \varepsilon y_1(x) + O(\varepsilon^2)$$

$$y(x) = B \sin(x) + \varepsilon (C \cos(x) + D \sin(x) + Bx \sin(x)) + O(\varepsilon^2)$$

Şimdi $y(0) = 0$ ve $y(1) = 1$ koşullarını uygulayalım

$$y(0) = B \sin(0) + \varepsilon (C \cos(0) + D \sin(0) + B(0) \sin(0)) + O(\varepsilon^2) = 0$$

$$\Rightarrow C = 0 \text{ dir.}$$

$$y(1) = B \sin(1) + \varepsilon (0 \cos(1) + D \sin(1) + B \sin(1)) + O(\varepsilon^2) = 1$$

$$1 = \sin(1) (B + \varepsilon (D + B))$$

$$B \sin(1) \approx 1$$

$$0 \Rightarrow D = B$$

$$B = \frac{1}{\sin(1)}$$

$$y(x) = \frac{1}{\sin(1)} \sin(x) + \varepsilon \left(\frac{1}{\sin(1)} \sin(x) - \frac{x}{\sin(1)} \sin(x) \right) + O(\varepsilon^2)$$

0-1 Bu sınır-değer probleminin tam çözümü bulunur ve $0 < \varepsilon \ll 1$ için asılının ilk iki terimini hesaplayın. Önce karakteristik denklemini yazalım.

$$r^2 + 2\varepsilon r + 1 = 0$$

$$r = \frac{-2\varepsilon \pm \sqrt{(2\varepsilon)^2 - 4}}{2} = -\varepsilon \pm \sqrt{-1 + \varepsilon^2} = -\varepsilon \pm \sqrt{1 - \varepsilon^2}$$

Böylelikle Genel çözüm

$$y(x) = e^{-\varepsilon x} (A \cos(\sqrt{1 - \varepsilon^2} x) + B \sin(\sqrt{1 - \varepsilon^2} x))$$

SK $\rightarrow y(0) = 0 \Rightarrow e^0 (A \cos(0) + B \sin(0)) = A \Rightarrow A = 0$

$\rightarrow y(1) = 1 \Rightarrow e^{-\varepsilon} (A \cos(\sqrt{1 - \varepsilon^2}) + B \sin(\sqrt{1 - \varepsilon^2})) \Rightarrow B = \frac{e^{\varepsilon}}{\sin(\sqrt{1 - \varepsilon^2})}$

şimdi $0 < \varepsilon \ll 1$ old için Taylor serisini kullanalım

$$* e^{\varepsilon} = 1 + \varepsilon + \frac{\varepsilon^2}{2} + O(\varepsilon^3)$$

$$* \sqrt{1 - \varepsilon^2} = 1 - \frac{\varepsilon^2}{2} + O(\varepsilon^4)$$

$$* \sin(\sqrt{1 - \varepsilon^2}) = \sin\left(1 - \frac{\varepsilon^2}{2}\right) \approx \sin(1) - \frac{\varepsilon^2}{2} \cos(1)$$

0 zaman

$$B \approx \frac{1 + \varepsilon}{\sin(1)}, \quad (\varepsilon \text{ daha küçük mertebesi})$$

$$* e^{-\varepsilon x} = 1 - \varepsilon x + O(\varepsilon^2)$$

$$* \sin\left(\left(1 - \frac{\varepsilon^2}{2}\right)x\right) = \sin(x) - \frac{\varepsilon^2}{2} x \cos(x) + O(\varepsilon^4)$$

} üstüne koyulur

$$y(x) \approx \frac{1 + \varepsilon}{\sin(1)} (1 - \varepsilon x) \left(\sin(x) - \frac{\varepsilon^2}{2} x \cos(x) \right)$$

$$= \frac{1}{\sin(1)} \left(\sin(x) + \varepsilon (1 - x) \sin(x) + O(\varepsilon^2) \right) \Rightarrow \text{(2) sıkkında}$$

asimptotik çözüm
yazılır

C-1 Bu problemi düzenli m. yavaş telat bir pertorasyon problemi m.
Eğer $\varepsilon \rightarrow 0$ limiti alınırken dif. denk. mertebesi
değişmiyorsa **Düzenli pertorasyon** diyorsa **telat pertorasyon**

$$\frac{d^2 y}{dx^2} + 2\varepsilon \frac{dy}{dx} + y = 0$$

Eğer $\varepsilon = 0$ koyarsak

$$\frac{d^2 y}{dx^2} + y = 0$$

Bu hden ikinci mertebeden bir dif denktir

- * Mertaba değişmez
- * Sınır tabaka oluşmaz
- * Çözüm denizinde ani değ. yde.

Düzenli
Pertorasyon

25000 $0 < \varepsilon < 1$ için given denkleminin as. dif. denk. sızınan

$$\varepsilon \frac{d^2 y}{dx^2} + y \frac{dy}{dx} - y^3 = 0, \quad y(0) = 0, \quad y(1) = 1, \quad 0 \leq x \leq 1$$

ve denklemin sızınan $x=0$ nok'tas'ında (tekl' bir sınır tabaka sızın)

2) Bu sızınan için yokes tekl' bir potansiyel problem

$\varepsilon \rightarrow 0$ limiti sızınan

$$y \frac{dy}{dx} - y^3 = 0$$

1. mertebeden bir dif. denk. den mertebe sızın

bl' öne' dı' sızınan $y(x) = \frac{1}{2-x}$ ol'du' gösteren ve $0 \leq x \leq 1$ için g'et'li' sızınan

Bu sızınan için $\varepsilon = 0$ limitini sızın

$$y \frac{dy}{dx} - y^3 = 0$$

$$y \left(\frac{dy}{dx} - y^2 \right) = 0, \quad y = 0 \text{ d'ezere}$$

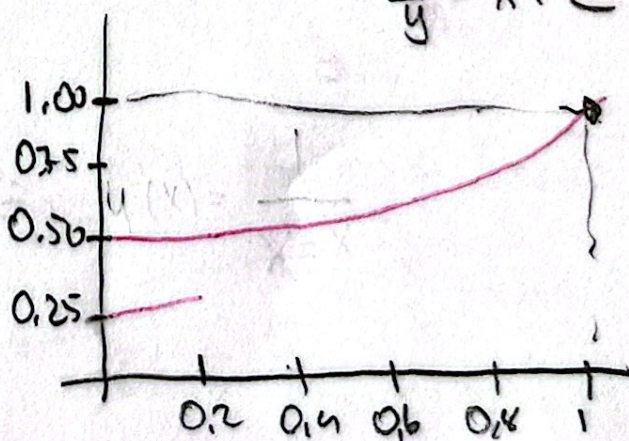
$$\frac{dy}{dx} = y^2 \Rightarrow \frac{dy}{y^2} = dx \Rightarrow \int \frac{1}{y^2} dy = \int dx$$

$$-\frac{1}{y} = x + C$$

$$\Rightarrow y(x) = \frac{-1}{x-C} = \frac{1}{C-x}$$

$$y(1) = \frac{1}{C-1} = 1 \Rightarrow C = 2$$

$$y(x) = \frac{1}{2-x}$$



C-1) Uygun ϵ değeri için $X = \frac{x}{\epsilon}$ old. gösterin ve buna karşın gelen ϵ cözümünü bul. $y = Y(X)$ bulunuz $X \geq 0$ den başlayarak

$$\frac{dy}{dx} = \frac{dY}{dX} \frac{dX}{dx} = \frac{dY}{dX} \epsilon$$

$$\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dY}{dX} \epsilon \right) = \frac{dX}{dx} \frac{d}{dX} \left(\frac{dY}{dX} \epsilon \right) = \epsilon^{-1} \left(\frac{d^2Y}{dX^2} \epsilon \right) = \epsilon^2 \left(\frac{d^2Y}{dX^2} \right)$$

$$\epsilon \left(\epsilon^2 \frac{d^2Y}{dX^2} \right) + Y \left(\epsilon^{-1} \frac{dY}{dX} \right) - Y^3 = 0$$

$$\frac{d^2Y}{dX^2} + Y \frac{dY}{dX} - \epsilon Y^3 = 0, \quad \epsilon \ll 1 \text{ old. için son term ihmal edilebilir.}$$

$$\frac{d^2Y}{dX^2} + Y \frac{dY}{dX} = 0, \quad \frac{dY}{dX} = P \text{ demek } \frac{d^2Y}{dX^2} = \frac{dP}{dY} \frac{dY}{dX} = P \frac{dP}{dY}$$

$$\Rightarrow P \frac{dP}{dY} + P Y = 0 \quad \Rightarrow \frac{dP}{dY} + Y = 0 \quad \left(\begin{array}{l} \text{Bernoulli denkleminin} \\ \text{lineer dif denkleminin} \end{array} \right)$$

$$\Rightarrow dP = -Y dY$$

$$\Rightarrow P = -\frac{Y^2}{2} + C_1$$

$$\Rightarrow \frac{dY}{dX} = -\frac{Y^2}{2} + C_1$$

$$\Rightarrow \frac{2dY}{2C_1 - Y^2} = dX, \quad 2C_1 = 2^2 \Rightarrow 2 = \sqrt{2C_1}$$

$$\Rightarrow 2 \int \frac{dY}{2^2 - Y^2} = \frac{2}{2} \tanh^{-1} \left(\frac{Y}{2} \right) \Rightarrow \frac{2}{2} \tanh^{-1} \left(\frac{Y}{2} \right) = X + C_2 \Rightarrow \frac{Y}{2} = \tanh \left(\frac{1}{2} (X + C_2) \right)$$

$$\Rightarrow Y = 2 \tanh \left(\frac{1}{2} (X + C_2) \right)$$

$$Y(0) = 2 \tanh\left(\frac{2}{2} C_2\right) = 0$$

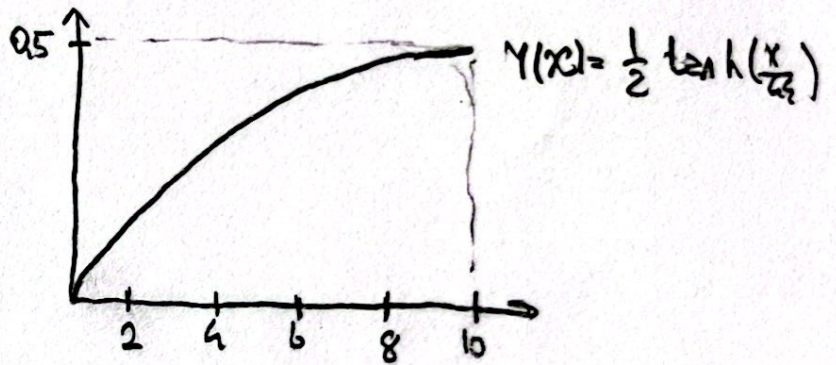
$$\tanh\left(\frac{2}{2} C_2\right) = 0 \Rightarrow \frac{2}{2} C_2 = 0 \Rightarrow C_2 = 0$$

$$Y(x) = 2 \tanh\left(\frac{2}{2} x\right) \quad \left. \begin{array}{l} x \rightarrow 0 \text{ gidenken } x \rightarrow \infty \text{ limitinde eğiliminde} \\ \cdot x \rightarrow \infty \text{ olduğunda } \tanh\left(\frac{2}{2} x\right) \rightarrow 1 \\ \cdot \text{ Dolayısıyla } Y \rightarrow 2 \\ \cdot x \rightarrow 0 \text{ için dış çözüm } y(x) \rightarrow \frac{1}{2} \end{array} \right\}$$

$$\text{Dolayısıyla } 2 = \frac{1}{2} \text{ sonucu olarak}$$

$$Y(x) = \frac{1}{2} \tanh\left(\frac{x}{4}\right)$$

$$Y(x) = \frac{1}{2} \tanh\left(\frac{x}{4\epsilon}\right)$$



$$d-1) y = \frac{1}{2} \tanh\left(\frac{x}{4\epsilon}\right) + \frac{x}{2(2-x)} \text{ birleştirilerek yazılabilir}$$

ikinci çözüm limiti ($x \rightarrow \infty$)

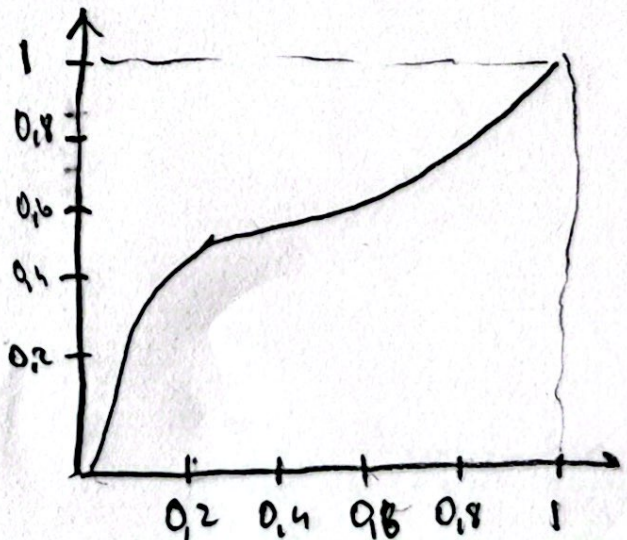
$$\frac{1}{2} \tanh\left(\frac{x}{4}\right) \rightarrow 1 \text{ dolayısıyla } y_{ic}(x) \rightarrow \frac{1}{2}$$

Dış çözüm limiti ($x \rightarrow 0$)

$$\lim_{x \rightarrow 0} \frac{1}{2-x} = \frac{1}{2}$$

$$\text{Birleştirilerek form.} = y(x) + y_{dış} - \text{birleştirilme} \\ = \frac{1}{2} \tanh\left(\frac{x}{4}\right) + \frac{1}{2-x} - \frac{1}{2}$$

$$Y(x) = \frac{1}{2} \tanh\left(\frac{x}{4}\right) + \frac{1}{2(2-x)}$$



3.9.11 $0 < \varepsilon \ll 1$ sein genau für ε diff. den asymptotischen

$$\varepsilon y'' + (4+x^2)(y'+2y) = 0, \quad 0 \leq x \leq 1$$

von der Form $x=0$ ist da ε klein sein ε ist klein
 2. Es ist asymptotisch genau $\varepsilon \ll 1$ ist ε klein sein ε ist klein
 Zahlen oder oder

$$\varepsilon y'', \text{ immer } \varepsilon \ll 1 \Rightarrow (4+x^2)(y'+2y) = 0$$

$$y'+2y=0 \Rightarrow \frac{dy}{dx} = -2y \Rightarrow \int \frac{dy}{y} = -2 \int dx \Rightarrow \ln y = -2x + C_0 \Rightarrow y_0(x) = C_0 e^{-2x}$$

$$x = \frac{x}{\varepsilon^\alpha}, \quad \alpha > 0 \quad \text{u.} \quad x=0, \quad \varepsilon \rightarrow 0, \quad y=Y(x) \quad 0 \leq x \leq 1$$

$$\frac{dy}{dx} = \frac{dY}{dX} \frac{dX}{dx} = \varepsilon^{-\alpha} \frac{dY}{dX}, \quad \frac{d^2 y}{dx^2} = \varepsilon^{-2\alpha} \frac{d^2 Y}{dX^2}$$

$$\varepsilon \varepsilon^{-2\alpha} \frac{d^2 Y}{dX^2} + (4 + \varepsilon^{2\alpha} X^2) \left(\varepsilon^{-\alpha} \frac{dY}{dX} + 2Y \right) = 0 \quad (*)$$

$$\varepsilon^{1-2\alpha} = \varepsilon^{-\alpha} \Rightarrow \alpha = 1 \Rightarrow x = \frac{x}{\varepsilon} \text{ ist}$$

$$(*) \text{ dazusetzen } \varepsilon^{-1} \frac{d^2 Y}{dX^2} + (4 + O(\varepsilon^2)) \left(\varepsilon^{-1} \frac{dY}{dX} + 2Y \right) = 0$$

$$\frac{d^2 Y}{dX^2} + 4 \frac{dY}{dX} + \varepsilon 8Y = 0 \Rightarrow \frac{d^2 Y}{dX^2} + 4 \frac{dY}{dX} = 0$$

Charakteristisches Polynom

$$r^2 + 4r = 0 \Rightarrow r = 0, r = -4$$

$$\text{Gemein } Y(x) \Rightarrow Y(X) = C_1 e^{0X} + C_2 e^{-4X} = C_1 + C_2 e^{-4X}$$

b) Van Dyke eigene Ansatz: y_2 ist durch Nullsetzen gegeben

Das System $x \rightarrow 0$ ist linear

$$y_{dy}(x) = C_0 e^{-2x} = C_0 (1 - 2x + 2x^2 + \dots)$$

In System $x \rightarrow \infty$ ist linear

$$y_{lg}(x) = C_1 + C_2 e^{-4x} \approx C_1$$

Eigene

$$C_0 = C_1 \Rightarrow y_{in}(x) = C_0 + C_2 e^{-4x}$$

$$\text{In System Torus} \Rightarrow \frac{dY}{dx} = \varepsilon^{-1} \frac{dY}{dX} = \varepsilon^{-1} 4C_2 e^{-4x} = 4C_2 \varepsilon^{-(4x+1)}$$
$$y'_{in}(x) = 4C_2 \varepsilon^{-(\frac{4x}{\varepsilon} + 1)}$$

$$\text{Das System Torus} \Rightarrow -2C_0 e^{-2x}$$

$$y'(x) = y'_{in} + y'_{lg} = -2C_0 e^{-2x} + 4C_2 \varepsilon^{-(\frac{4x}{\varepsilon} + 1)}$$

$$y'(0) = -2C_0 - 4 \frac{C_2}{\varepsilon} =$$

$$y'(0) \sim \frac{4\varepsilon^2}{\varepsilon} - 4\varepsilon^2 \tan^{-1}\left(\frac{1}{2}\right) \Rightarrow C_0 = 2\varepsilon^2 - 4\varepsilon^2 \tan^{-1}\left(\frac{1}{2}\right), C_2 = -\varepsilon^2$$

2-1) O(ε)'e werden gewollt oder bei beliebiger Ordnung beliebig

$$y_{gesamt} = y_{dy}(x) + y_{lg}(x) - y_{eg}(x)$$

$$= C_0 e^{-2x} + C_0 + C_2 e^{-\frac{4x}{\varepsilon}} - C_0$$

$$= C_0 e^{-2x} + C_2 e^{-\frac{4x}{\varepsilon}}$$

4.500 $\varepsilon y'' + x^{\frac{1}{2}} y' + y = 0$, $y(0) = 0$, $y(1) = 1$, $0 \leq x \leq 1$ diň deňleşmisi.
 Eneň is, ne diň, eoz $0 < \varepsilon \ll 1$ iň belarlyn.

Suň tabarka $x=0$ nolabunda

$\varepsilon \rightarrow 0$ ol iňio $\varepsilon y''$ ihad edilä.

$$x^{\frac{1}{2}} y' + y = 0 \Rightarrow y' + x^{-\frac{1}{2}} y = 0, \quad \mu(x) = e^{\int \frac{1}{x^{\frac{1}{2}}} dx} = e^{2x^{\frac{1}{2}}}$$

$$\Rightarrow \frac{d}{dx} (y e^{2x^{\frac{1}{2}}}) = 0 \Rightarrow y_{d1s}(x) = C e^{-2x^{\frac{1}{2}}}$$

$$S.V. \rightarrow y(0) = 0 \Rightarrow y_{d1s}(0) = C \neq 0$$

$$\rightarrow y(1) = 1 \Rightarrow y_{d1s}(1) = C e^{-2} \Rightarrow C = e^2, \quad y_{d1s}(x) = e^2 e^{-2x^{\frac{1}{2}}}$$

$$x = \varepsilon^{\frac{2}{3}} \chi, \quad y(x) = Y(\chi), \quad \chi = 0 \text{ iň}, \quad \frac{dy}{dx} = \varepsilon^{-\frac{1}{3}} \frac{dY}{d\chi}, \quad \frac{d^2 y}{dx^2} = \varepsilon^{-\frac{2}{3}} \frac{d^2 Y}{d\chi^2}$$

$$\varepsilon \cdot \varepsilon^{-\frac{2}{3}} \frac{d^2 Y}{d\chi^2} + (\varepsilon^{\frac{1}{2}} \varepsilon^{-\frac{1}{3}}) \frac{dY}{d\chi} + Y = 0$$

$$\underbrace{O(\varepsilon^{\frac{1}{3}})}_{0} \underbrace{O(\varepsilon^{\frac{1}{6}})}_{0} \underbrace{O(1)}_{0} \Rightarrow 1 - 2\alpha = \frac{1}{3} \Rightarrow \alpha = \frac{2}{3}$$

$$\frac{d^2 Y}{d\chi^2} + \chi^{\frac{1}{2}} \frac{dY}{d\chi} + \varepsilon^{\frac{1}{3}} Y = 0 \Rightarrow \varepsilon y \text{ ihad edilä.}$$

$$\frac{d^2 Y}{d\chi^2} + \chi^{\frac{1}{2}} \frac{dY}{d\chi} = 0, \quad \mu(\chi) = e^{\int \chi^{\frac{1}{2}} d\chi} = e^{\frac{2}{3} \chi^{\frac{3}{2}}}$$

$$\Rightarrow \frac{d}{d\chi} (Y' e^{\frac{2}{3} \chi^{\frac{3}{2}}}) = 0 \Rightarrow Y'_{d1s} = C_1 e^{-\frac{2}{3} \chi^{\frac{3}{2}}}, \quad Y'_{d1s}(x) = C_1 \int e^{-\frac{2}{3} x^{\frac{3}{2}}} dx$$

1. Exkurs

Die Lösung $y(1) = e^2 e^{-2 \cdot 1^{\frac{1}{2}}} = e^2 e^{-2 \cdot \frac{1}{3} \cdot 1^{\frac{1}{2}}} \approx e^2 (1 - 2 \cdot \frac{1}{3} \cdot 1^{\frac{1}{2}} + \dots)$

1a. Lösung: $x \rightarrow \infty$, $\lim$

$$Y(x) \rightarrow C_1 \int_0^{\infty} e^{-\frac{2}{3} s^{\frac{3}{2}}} ds$$

3. Integral

$$\int_0^{\infty} e^{-a x^p} dx = \frac{1}{p} (a)^{-\frac{1}{p}} \Gamma\left(\frac{1}{p}\right), \quad p = \frac{3}{2}, \quad a = \frac{2}{3} \Rightarrow \frac{2}{3} \left(\frac{2}{3}\right)^{-\frac{2}{3}} \Gamma\left(\frac{2}{3}\right) \\ \Rightarrow \left(\frac{2}{3}\right)^{\frac{1}{3}} \Gamma\left(\frac{2}{3}\right)$$

$$Y(\infty) = C_1 \left(\frac{2}{3}\right)^{\frac{1}{3}} \Gamma\left(\frac{2}{3}\right) \approx e^2 \Rightarrow C_1 = \left(\frac{3}{2}\right)^{\frac{1}{3}} \frac{e^2}{\Gamma\left(\frac{2}{3}\right)}$$

$$y'(x) = Y'\left(\frac{x^3}{3}\right) = C_1 e^{-\frac{2}{3} \left(\frac{x^3}{3}\right)^{\frac{3}{2}}}$$

$$y'(0) = Y'(0) = C_1 e^{-\frac{2}{3}} = e^{-\frac{2}{3}} \frac{e^2}{\Gamma\left(\frac{2}{3}\right)} \left(\frac{3}{2}\right)^{\frac{1}{3}}$$

$$\Gamma(1-z) \Gamma(z) = \frac{\pi}{\sin(\pi z)} \quad \text{für } z = \frac{2}{3}$$

5- $2\varepsilon y' + e^x y' - y = 0$, $y(0)=0$, $y(1)=1$ $0 \leq x \leq 1$ Sınır tabaka $x=0$

$\varepsilon \rightarrow 0$ sınırında en yavaşla tanımlı form yde sayıldaki

gelişimin

$$e^x y' - y = 0 \Rightarrow \int \frac{dy}{y} = \int \frac{dx}{e^x} \Rightarrow \ln y = -e^x + C \Rightarrow y = e^{(-e^x + C)} = A e^{-e^x} \Rightarrow y_0 = e^{-e^x}$$

$y(0)=0$ koşulu sağlanmaz, sınır tabaka $x=0$ vardır ve burada da sınır gerilimi

$$x = \varepsilon^2 \chi, \quad y(x) = Y(\chi), \quad \frac{dy}{dx} = \varepsilon^{-2} \frac{dY}{d\chi}, \quad \frac{d^2 y}{dx^2} = \varepsilon^{-4} \frac{d^2 Y}{d\chi^2}$$

$$2\varepsilon \varepsilon^{-2} \frac{d^2 Y}{d\chi^2} + (1 + \varepsilon \chi) \varepsilon^{-2} \frac{dY}{d\chi} - Y = 0$$

$$O(\varepsilon^{-2}) \quad O(\varepsilon^{-1}) \quad O(1) \Rightarrow 1 - 2\chi = -\chi \Rightarrow \chi = 1 \Rightarrow x = \varepsilon^2 \chi$$

$$2\varepsilon^{-1} \frac{d^2 Y}{d\chi^2} + \varepsilon^{-1} \frac{dY}{d\chi} + \chi \frac{dY}{d\chi} - Y = 0$$

$$2 \frac{d^2 Y}{d\chi^2} + \frac{dY}{d\chi} + \chi \frac{dY}{d\chi} - \varepsilon Y = 0, \quad \varepsilon \rightarrow 0 \text{ için } e^{\varepsilon \chi} \approx 1, \quad \varepsilon Y \approx 0$$

$$2 \frac{d^2 Y}{d\chi^2} + \frac{dY}{d\chi} = 0 \quad \text{Karakteristik denkleme} \quad 2r^2 + r = 0 \Rightarrow r = 0, r = -\frac{1}{2} \Rightarrow Y(\chi) = C_1 + C_2 e^{-\frac{\chi}{2}}$$

$$y(0) = Y(0) = C_1 + C_2 = 0 \Rightarrow C_1 = -C_2 \Rightarrow Y(\chi) = C_2 (e^{-\frac{\chi}{2}} - 1)$$

$$y(1) = C_2 (e^{-\frac{1}{2\varepsilon^2}} - 1)$$

$$y(1) = C_2 (e^{-\frac{1}{2\varepsilon^2}} - 1) = 1 \Rightarrow C_2 = \frac{1}{e^{-\frac{1}{2\varepsilon^2}} - 1}$$

$$\text{Erlene } \chi \rightarrow \infty \Rightarrow e^{-\chi} \rightarrow 0 \Rightarrow y_{1s} = -C_2 \quad \left. \begin{array}{l} \chi \rightarrow 0 \Rightarrow e^{\chi} \rightarrow 1 \Rightarrow y_{0s} = -e^e \end{array} \right\} \Rightarrow C_2 = -e^e$$

$$\chi \rightarrow 0 \Rightarrow e^{\chi} \rightarrow 1 \Rightarrow y_{0s} = -e^e$$

$$y_{bilgi} = e^{-e^x} - e^e (e^{-\frac{x}{2\varepsilon^2}} - 1) + e^e \Rightarrow A = \frac{1}{e^{-e^e} - 1}$$

299/11 $4\varepsilon y'' + (x^2 + 1)y' - xy = 0$, $y(0) = 1$, $y(1) = 1$, $0 \leq x \leq 1$ Sturm-Liouville
 $x=0$

$\varepsilon \rightarrow 0$ dimeret $4\varepsilon y''$ ihad walar

$$(x^2 + 1)y' - xy = 0 \Rightarrow y' = \frac{xy}{(x^2 + 1)} \Rightarrow \int \frac{dy}{y} = \int \frac{x}{x^2 + 1} dx \Rightarrow \ln y = \frac{1}{2} \ln(x^2 + 1) + C \Rightarrow y = A \sqrt{x^2 + 1}$$

$x = \varepsilon^{\frac{1}{2}} \chi$, $y(x) = Y(\chi)$, $\frac{dy}{dx} = \varepsilon^{\frac{1}{2}} \frac{dY}{d\chi}$, $\frac{d^2y}{dx^2} = \varepsilon^{\frac{1}{2}} \frac{d^2Y}{d\chi^2}$ $y = \frac{1}{\sqrt{2}} \sqrt{x^2 + 1}$

$$4\varepsilon \left(\varepsilon^{\frac{1}{2}} \frac{d^2Y}{d\chi^2} \right) + (\varepsilon^2 \chi^2 + 1) \varepsilon^{\frac{1}{2}} \frac{dY}{d\chi} - \varepsilon^{\frac{1}{2}} \chi Y = 0$$

$O(\varepsilon^{\frac{1}{2}})$

$O(\varepsilon^{\frac{1}{2}})$

$\varepsilon^{\frac{1}{2}} \Rightarrow 1 - 2\varepsilon^{\frac{1}{2}} + \dots \Rightarrow \alpha = 1 - \varepsilon^{\frac{1}{2}}$

$$4 \frac{d^2Y}{d\chi^2} + (\chi^2 + 1) \frac{dY}{d\chi} - \chi Y = 0$$

$$4Y'' + Y' + \chi^2 Y' + \chi^2 Y = 0$$

$$4Y'' + Y' = 0 \Rightarrow r(r+1) = 0 \Rightarrow Y_{in} = C_1 + C_2 e^{-\frac{\chi}{4}}$$

$$y(0) = 1 \Rightarrow Y(0) = C_1 + C_2 = 1 \Rightarrow C_1 = C_2 \Rightarrow Y(\chi) = C_1 (1 + e^{-\frac{\chi}{4}})$$

$$y(1) = 1 \Rightarrow Y(\varepsilon^{-\frac{1}{2}}) = C_1 (1 + e^{-\frac{\varepsilon^{-\frac{1}{2}}}{4}}) = 1 \Rightarrow C_1 = \frac{1}{1 + e^{-\frac{\varepsilon^{-\frac{1}{2}}}{4}}}$$

$\chi \rightarrow \infty \Rightarrow e^{-\chi} \rightarrow 0 \Rightarrow C_1$ } $C_1 = \frac{1}{\sqrt{2}}$

$\chi \rightarrow 0 \Rightarrow \frac{1}{\sqrt{2}} \sqrt{x^2 + 1} = \frac{1}{\sqrt{2}}$

$y_{block} = y_{d,s} + y_{i,s} = c_1 y_{s,n}$

$$\frac{1}{\sqrt{2}} \sqrt{x^2 + 1} + \frac{1}{\sqrt{2}} (1 + e^{-\frac{x}{4}}) = \frac{1}{\sqrt{2}}$$

Öğr. $\varepsilon y'' + (x+2)y' + y = 1$, $y(0)=0$, $y(1)=2$
 y'' in katsayı $\varepsilon \ll 1 \rightarrow$ sınır tabakası olabilir
 y' in katsayısı $x+2 > 0$ yani $x=0$ 'da
 Aşağıdaki $\varepsilon \rightarrow 0$ yani $\varepsilon y''$ ihmal edilir.

$$\frac{d}{dx}[u \cdot v] = u' \cdot v + u \cdot v'$$

$$(x+2)y' + y = 1 \Rightarrow y + \frac{1}{x+2}y = \frac{1}{x+2}, \quad \mu(x) = e^{\int \frac{1}{x+2} dx} = e^{\ln(x+2)} = x+2$$

$$\Rightarrow \frac{d}{dx}[(x+2)y] = 1 \Rightarrow (x+2)y = x + C \Rightarrow y_{d1} = \frac{x+C}{x+2}$$

$$x = \varepsilon^2 \chi, \quad y(x) = Y(\chi), \quad \frac{dy}{dx} = \varepsilon^{-2} \frac{dY}{d\chi}, \quad \frac{d^2y}{dx^2} = \varepsilon^{-2} \frac{d^2Y}{d\chi^2}$$

$$\varepsilon \varepsilon^{-2} \frac{d^2Y}{d\chi^2} + \varepsilon^2 \chi \cdot \varepsilon^{-2} \frac{dY}{d\chi} + 2 \varepsilon^{-2} \frac{dY}{d\chi} + Y = 1$$

$$\varepsilon^{-1} \frac{d^2Y}{d\chi^2} + \chi \frac{dY}{d\chi} + 2 \frac{dY}{d\chi} + Y = \varepsilon$$

$\varepsilon \ll 1 \Rightarrow \chi = \varepsilon \chi$

$$\frac{d^2Y}{d\chi^2} + 2 \frac{dY}{d\chi} = 0 \Rightarrow Y_{ic} = A + B e^{-2\chi}$$

$$x=0 \Rightarrow \chi=0 = Y(0) = A+B=0 \Rightarrow A=-B \Rightarrow Y_{ic}(\chi) = A(1 - e^{-2\chi})$$

$$\left. \begin{array}{l} \chi \gg 0 \quad y_{d1} \Rightarrow \frac{\varepsilon}{2} \\ \chi \gg 0 \quad y_{ic} \Rightarrow A \end{array} \right\} A = \frac{\varepsilon}{2} \Rightarrow$$

$$Y_{Bilgi} = Y_{d1} + Y_{ic} - Y_{ortak}$$

$$= \frac{x+C}{x+2} + \frac{\varepsilon}{2}(1 - e^{-2\frac{x}{\varepsilon}}) - \frac{\varepsilon}{2}$$

$$y(1) \approx \frac{1+C}{2} = 2 \Rightarrow C=5$$

$$y(x) = \frac{x+5}{x+2} - \frac{5e^{-2\frac{x}{\varepsilon}}}{2}$$

$\epsilon y'' - (x+2)y' + y = 1, y(1)=0, y(1)=2 \quad 0 < \epsilon \ll 1$ iain

$P(x) = -(x+2) < 0$ d'd' d' d' $x=1, \epsilon \rightarrow 0 \Rightarrow y''$ phthal d' d' d'

$-(x+2)y' + y = 1 \Rightarrow y' - \frac{y}{(x+2)} = \frac{1}{(x+2)} \Rightarrow \mu(x) = e^{-\int \frac{1}{x+2}} = e^{-\ln(x+2)} = \frac{1}{x+2}$

$\Rightarrow \frac{y'}{x+2} - \frac{y}{(x+2)^2} = \frac{1}{(x+2)^2} \Rightarrow \frac{d}{dx} \left(\frac{y}{x+2} \right) = \frac{1}{(x+2)^2} \Rightarrow \frac{y}{x+2} = \int \frac{1}{(x+2)^2} dx \Rightarrow y = 1 + C(x+2)$
 $y = 1 - \frac{x+2}{2}$

$x = 1 - \epsilon^\alpha x, y(x) = Y(x), \frac{dy}{dx} = -\epsilon^\alpha \frac{dY}{dx}, \frac{d^2y}{dx^2} = \epsilon^{-2\alpha} \frac{d^2Y}{dx^2}$

$\epsilon \epsilon^{-2\alpha} \frac{d^2Y}{dx^2} - (3 - \epsilon^\alpha x) \left(-\epsilon^\alpha \frac{dY}{dx} \right) + Y = 1$
 $O(\epsilon^{1-2\alpha}) \quad O(\epsilon^\alpha)$

$\Rightarrow 1-2\alpha = 2 \Rightarrow \alpha = \frac{1}{3} \Rightarrow x = 1 - \epsilon^{\frac{1}{3}} x$

$\epsilon^{\frac{1}{3}} Y'' + 3\epsilon^{\frac{1}{3}} Y' - xY' + Y = 1$

$Y'' + 3Y' + \epsilon^{\frac{1}{3}} Y' - \epsilon^{\frac{1}{3}} Y = \epsilon^{\frac{1}{3}} \Rightarrow \epsilon \rightarrow 0 =$ iain

$Y'' + 3Y' = 0 \Rightarrow r^2 + 3r = 0, r = 0, r = -3 \Rightarrow Y(x) = A + Be^{-3x}$

$\bullet x=1 \Rightarrow x=0 \Rightarrow Y(0) = A + Be^0 = 2 \Rightarrow A+B=2$

di eozon $x \rightarrow 0$ iain

$y_{d' d' d'}(x) = 1 + \frac{x+2}{2} = -\frac{1}{2} 3e$

in eozon $x \rightarrow \infty$ iain

$y_{ic}(x) = A + Be^\infty = A$

$\left. \begin{array}{l} A = -\frac{1}{2} \\ B = \frac{5}{2} \end{array} \right\}$

Biksit

$y_{biksit} = 1 - \frac{x+2}{2} - \frac{1}{2} + \frac{5}{2} e^{-3 \left(\frac{1-x}{\epsilon^{\frac{1}{3}}} \right)} - \left(-\frac{1}{2} \right) = 1$

$y(0) = 1 + 2C = 0 \Rightarrow \frac{5}{2} e^{-\frac{3 \left(\frac{1-x}{\epsilon^{\frac{1}{3}}} \right)}{2}} = \frac{1}{2} \Rightarrow -\frac{1}{2} + B = 2 \Rightarrow B = \frac{5}{2}$

$y_{biksit} = 1 - \frac{x+2}{2} + \frac{5}{2} e^{-\frac{3}{1-\epsilon^{\frac{1}{3}}}}$

$$\epsilon y'' + y' - e^y = 0, \quad y(0), y(1) = \ln 3, \quad 0 < \epsilon \ll 1$$

$P(x)=1 \rightarrow$ solit ve pozitif y'ni soldan sağa taşıyoruz $x=0$ den itibaren

$$\epsilon \rightarrow 0 \Rightarrow y'' \text{ birim ihmal ediliyor. } \{-\ln(-1-\epsilon) = \ln 3 \Rightarrow \frac{1}{-1-\epsilon} = 3 \Rightarrow \epsilon = -\frac{4}{3}\}$$

$$y' - e^y = 0 \Rightarrow \frac{dy}{dx} = e^y \Rightarrow \int \frac{dy}{e^y} = \int dx \Rightarrow -e^{-y} = x + C \Rightarrow -y = \ln(x+C) \Rightarrow y = -\ln(x+C)$$

$$x = \epsilon^{\frac{1}{2}} \chi, \quad y(x) = Y(\chi), \quad \frac{dy}{dx} = \epsilon^{-\frac{1}{2}} \frac{dY}{d\chi}, \quad \frac{d^2y}{dx^2} = \epsilon^{-\frac{3}{2}} \frac{d^2Y}{d\chi^2}$$

$$\underbrace{\epsilon \epsilon^{-\frac{3}{2}} \frac{d^2Y}{d\chi^2}}_{O(\epsilon^{-\frac{1}{2}})} + \underbrace{\epsilon^{-\frac{1}{2}} \frac{dY}{d\chi}}_{O(\epsilon^{-\frac{1}{2}})} = e^Y = 0 \Rightarrow 1-2\alpha = -2 \Rightarrow \alpha = 1, \chi = \epsilon x$$

$$\frac{d^2Y}{d\chi^2} + \frac{dY}{d\chi} - \epsilon^{\frac{1}{2}} e^Y = 0 \Rightarrow \epsilon \rightarrow 0 \text{ için}$$

$$r^2 + r = 0 \Rightarrow r = 0, r = -1 \Rightarrow Y(\chi) = A + B e^{-\chi}$$

Esaslar $x \rightarrow 0$

bu bölge için y'lerin bölgesi ($x \rightarrow 0$)

$$y(x) \rightarrow -\ln(-\frac{4}{3})$$

1. bölge için y'lerin bölgesi ($x \rightarrow \infty$)

$$Y(\chi) \rightarrow -B$$

$$Y(0) = A + B e^0 = 0 \Rightarrow A = -B$$

$$Y(\chi) = B(e^{-\chi} - 1)$$

$$-B = \ln(-\frac{4}{3}) \Rightarrow -\frac{4}{3} = e^{-B}$$

Bölge A'ları

$$y_{\text{bölge A}} = -\ln(-x - \frac{4}{3}) + (\ln(-\frac{4}{3}))(e^{-\frac{x}{\epsilon}} - 1) - \ln(-\frac{4}{3})$$

$$y(1) \approx -\ln(-1-\epsilon) = \ln 3 \Rightarrow -\frac{4}{3} = e^{-\frac{1}{\epsilon}}$$

$$y_{\text{bölge B}} = -\ln(-x - \frac{4}{3}) - \ln(-\frac{4}{3}) \cdot (e^{-\frac{x}{\epsilon}})$$

$$\epsilon y'' + x^2 y' - y = 0, \quad y(0) = 1, \quad y(1) = 1, \quad 0 < \epsilon \ll 1 \text{ sein}$$

$x^2 > 0$ o.d. $x=0$ sein. Folglich, $\epsilon \rightarrow 0$, y' istel edlen

$$x^2 y' - y = 0 \Rightarrow x^2 \frac{dy}{dx} = y \Rightarrow \int \frac{dy}{y} = \int x^{-2} dx \Rightarrow \ln y = -x^{-1} + c \Rightarrow y = A e^{-x^{-1}} = \frac{A e^{-\frac{1}{x}}}{e^{-\frac{1}{x}}}$$

$y(1)=1 \Rightarrow e e^{-\frac{1}{x}}$

$$x = x \epsilon^\alpha, \quad y(x) = Y(X), \quad \frac{dy}{dx} = \epsilon^{-\alpha} \frac{dY}{dX}, \quad \frac{d^2 y}{dx^2} = \epsilon^{-2\alpha} \frac{d^2 Y}{dX^2}$$

$$\underbrace{\epsilon \epsilon^{-2\alpha} \frac{d^2 Y}{dX^2}}_{O(\epsilon^{1-2\alpha})} + \underbrace{x^2 \epsilon^{2\alpha} \epsilon^{-\alpha} \frac{dY}{dX}}_{O(\epsilon^\alpha)} - Y = 0$$

$$1-2\alpha = \alpha \Rightarrow \alpha = \frac{1}{3} \Rightarrow x = x \epsilon^{\frac{1}{3}}$$

$$\frac{d^2 Y}{dX^2} + X^2 \frac{dY}{dX} - Y = 0, \quad \epsilon \rightarrow 0 \quad \text{u.d.} \quad \frac{dY}{dX} = 0 \quad \text{o.d.}$$

$$Y' + X^2 Y = 0 \Rightarrow \int \frac{dY}{Y} = - \int X^2 dX \Rightarrow \ln |Y| = -\frac{X^3}{3} + c_1 = c_2 e^{-\frac{X^3}{3}}$$

$$Y'(X) = c_2 e^{-\frac{X^3}{3}} \Rightarrow Y(X) = c_2 \int_0^X e^{-\frac{s^3}{3}} ds + c_3$$

$$x=0 \Rightarrow X=0 \Rightarrow Y(0) = c_2 \cdot 0 + c_3 = 1 \Rightarrow c_3 = 1$$

Ueher limit ($X \rightarrow \infty$) $\Rightarrow$ Y ist konstant

$$c_2 \int_0^\infty e^{-\frac{s^3}{3}} ds + 1 = 1 \Rightarrow \lim_{X \rightarrow \infty} Y(X) = 1 + c_2 I$$

Dies wenn ($X \rightarrow 0$)

$$y_{\text{sub}}(x) = A e^{-\frac{1}{x}} \Rightarrow \lim_{x \rightarrow 0^+} y_{\text{sub}}(x) = 0$$

$$y_{\text{sub}} = e^{-\frac{1}{x}} + 1 - \frac{1}{I} \int_0^{\frac{x}{\epsilon^{\frac{1}{3}}}} e^{-\frac{s^3}{3}} ds, \quad y(1) = e^{-1} + 1 - \frac{1}{I}$$

$$1 + c_2 I = 0$$

$$c_2 = -\frac{1}{I}$$

$$\epsilon y' + x^p y' - (p-1)y = 0, y(0)=1, y(\infty)=1, p>1$$

$x^p > 0$ için sınır tabakası $x=0$, $\epsilon \rightarrow 0$, y'' terimi yok sayılır

$$x^p y' - (p-1)y = 0 \Rightarrow \frac{dy}{dx} = \frac{(p-1)y}{x^p} \Rightarrow \int \frac{dy}{y} = \int \frac{p-1}{x^p} dx \Rightarrow \ln y = (p-1) \frac{x^{1-p}}{1-p} + C_1$$

$$y_{ds}(x) = A e^{-x^{1-p}} \Rightarrow y(0)=1 \Rightarrow A=1 \Rightarrow y_{ds}(x) = e^{1-x^{1-p}}$$

$$x = \epsilon^\alpha \chi, y(x) = Y(\chi)$$

$$\epsilon \epsilon^{-2\alpha} \frac{\partial^2 Y}{\partial \chi^2} + \epsilon^{1-p-\alpha} \chi^p \frac{\partial Y}{\partial \chi} - (p-1)Y = 0$$

$$\underbrace{O(\epsilon^{1-2\alpha})}_{0} + \underbrace{O(\epsilon^{1-p-\alpha})}_{0} \Rightarrow 1-2\alpha = -(p-1) \Rightarrow \alpha = \frac{1}{p+1}, x = \epsilon^{\frac{1}{p+1}} \chi$$

$$\epsilon^{\frac{p}{p+1}} (Y'' + \chi^p Y') - (p-1)Y = 0$$

$$Y'' + \chi^p Y' - (p-1)Y = 0, \epsilon \rightarrow 0 \text{ u } Y' = u \text{ için}$$

$$u' + \chi^p u = 0 \Rightarrow \int \frac{du}{u} = - \int \chi^p d\chi \Rightarrow \ln u = -\frac{\chi^{p+1}}{p+1} + C_2 \Rightarrow u = A e^{-\frac{\chi^{p+1}}{p+1}}$$

$$Y'(\chi) = A e^{-\frac{\chi^{p+1}}{p+1}} \Rightarrow Y(\chi) = A \int_0^\chi e^{-\frac{s^{p+1}}{p+1}} ds + C_3, Y(0)=1 \Rightarrow \underline{C_3=1}$$

$x \rightarrow 0, \chi \rightarrow \infty$ için

$$\left. \begin{array}{l} \lim_{\chi \rightarrow \infty} Y(\chi) = 1 + C_2 I \\ \lim_{x \rightarrow 0} y_{ds}(x) = e^{1-x} \Rightarrow 0 \end{array} \right\} 1 + C_2 I = 0 \Rightarrow C_2 = -\frac{1}{I}$$

Bilgi Aklın

$$y(x) = e^{1-x^{1-p}} + 1 - \frac{1}{I} \int_0^{\frac{x}{\epsilon^{\frac{1}{p+1}}}} e^{-\frac{s^{p+1}}{p+1}} ds, I = \int_0^\infty e^{-\frac{s^{p+1}}{p+1}} ds$$